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# ASSIGNMENT 8 - PYTHON BASICS AND DATA STRUCTURES
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# CLASS 1 ASSIGNMENT
# Question 1 - BMI CALCULATOR
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```
weight = float(input('Enter your weight in Kilograms(Kg): '))
height = float(input('Enter your height in Meters(M): '))

bmi = round(weight/height**2,2)

msg = '''
Body Mass Index Chart:
Between 18 - 25 = Normal Weight
Less than (<) 18 = Underweight
Greater than (>) 25 = Overweight OR Obese
'''

if bmi < 18:
    print(f'\nYour body mass index is {bmi}. You are Underweight!')
elif bmi >= 18 and bmi <= 25:
    print(f'\nYour body mass index is {bmi}. You are Normal Weight!')
else:
    print(f'\nYour body mass index is {bmi}. You are Overweight OR Obese!')

print(msg)
```

```
Enter your weight in Kilograms(Kg): 61
Enter your height in Meters(M): 1.52
```

Your body mass index is 26.4. You are Overweight OR Obese!

```
Body Mass Index Chart:
Between 18 - 25 = Normal Weight
Less than (<) 18 = Underweight
Greater than (>) 25 = Overweight OR Obese
```

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# Question 2 - Civil Servant Retirement Year Calculator
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```
title = input('Title: ')
first_name = input('First Name: ')
last_name = input('Last Name: ')

commencement_year = int(input('Year Commenced Civil Service: '))
birth_year = int(input('Birth Year: '))
current_year = int(input('Current Year: '))

current_age = current_year - birth_year
years_in_service = current_year - commencement_year

retirement_age = 60
retirement_year = birth_year + retirement_age
years_left = retirement_age - current_age

print(f'\nHello {title} {first_name} {last_name}.')
print(f'\nYou are {current_age} years old and you will be retiring in the year {retirement_year} at the age of {retirement_age}')
print(f'\nYou have spent {years_in_service} years of active service. You have {years_left} more years of active service left')
print(f'\nThank you for your service.')
```

```
Title: Mr.
First Name: Godbless
Last Name: Ejata
Year Commenced Civil Service: 2010
Birth Year: 1982
Current Year: 2026
```

Hello Mr. Godbless Ejata.

You are 44 years old and you will be retiring in the year 2042 at the age of 60.

You have spent 16 years of active service. You have 16 more years of active service left

Thank you for your service.

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africa = {
# WEST AFRICA
'Nigeria': {'Capital': 'Abuja', 'GDP': '477 billion USD', 'Region': 'West Africa'},
'Ghana': {'Capital': 'Accra', 'GDP': '76 billion USD', 'Region': 'West Africa'},
'Senegal': {'Capital': 'Dakar', 'GDP': '31 billion USD', 'Region': 'West Africa'},
'Ivory Coast': {'Capital': 'Yamoussoukro', 'GDP': '70 billion USD', 'Region': 'West Africa'},
'Mali': {'Capital': 'Bamako', 'GDP': '19 billion USD', 'Region': 'West Africa'},
'Burkina Faso': {'Capital': 'Ouagadougou', 'GDP': '18 billion USD', 'Region': 'West Africa'},
'Niger': {'Capital': 'Niamey', 'GDP': '14 billion USD', 'Region': 'West Africa'},
'Guinea': {'Capital': 'Conakry', 'GDP': '20 billion USD', 'Region': 'West Africa'},
'Benin': {'Capital': 'Porto-Novvo', 'GDP': '17 billion USD', 'Region': 'West Africa'},
'Togo': {'Capital': 'Lome', 'GDP': '9 billion USD', 'Region': 'West Africa'},
'Sierra Leone': {'Capital': 'Freetown', 'GDP': '4 billion USD', 'Region': 'West Africa'},
'Liberia': {'Capital': 'Monrovia', 'GDP': '4 billion USD', 'Region': 'West Africa'},
'Guinea-Bissau': {'Capital': 'Bissau', 'GDP': '2 billion USD', 'Region': 'West Africa'},
'Gambia': {'Capital': 'Banjul', 'GDP': '2 billion USD', 'Region': 'West Africa'},
'Cape Verde': {'Capital': 'Praia', 'GDP': '2 billion USD', 'Region': 'West Africa'},
'Mauritania': {'Capital': 'Nouakchott', 'GDP': '10 billion USD', 'Region': 'West Africa\n'},

# EAST AFRICA
'Kenya': {'Capital': 'Nairobi', 'GDP': '118 billion USD', 'Region': 'East Africa'},
'Ethiopia': {'Capital': 'Addis Ababa', 'GDP': '156 billion USD', 'Region': 'East Africa'},
'Tanzania': {'Capital': 'Dodoma', 'GDP': '79 billion USD', 'Region': 'East Africa'},
'Uganda': {'Capital': 'Kampala', 'GDP': '49 billion USD', 'Region': 'East Africa'},
'Rwanda': {'Capital': 'Kigali', 'GDP': '14 billion USD', 'Region': 'East Africa'},
'Somalia': {'Capital': 'Mogadishu', 'GDP': '8 billion USD', 'Region': 'East Africa'},
'Burundi': {'Capital': 'Gitega', 'GDP': '3 billion USD', 'Region': 'East Africa'},
'Djibouti': {'Capital': 'Djibouti City', 'GDP': '4 billion USD', 'Region': 'East Africa'},
'Eritrea': {'Capital': 'Asmara', 'GDP': '2 billion USD', 'Region': 'East Africa'},
'South Sudan': {'Capital': 'Juba', 'GDP': '11 billion USD', 'Region': 'East Africa'},
'Madagascar': {'Capital': 'Antananarivo', 'GDP': '14 billion USD', 'Region': 'East Africa'},
'Comoros': {'Capital': 'Moroni', 'GDP': '1 billion USD', 'Region': 'East Africa'},
'Mauritius': {'Capital': 'Port Louis', 'GDP': '14 billion USD', 'Region': 'East Africa'},
'Seychelles': {'Capital': 'Victoria', 'GDP': '2 billion USD', 'Region': 'East Africa\n'},

# SOUTHERN AFRICA
'South Africa': {'Capital': 'Pretoria', 'GDP': '405 billion USD', 'Region': 'Southern Africa'},
'Zimbabwe': {'Capital': 'Harare', 'GDP': '26 billion USD', 'Region': 'Southern Africa'},
'Zambia': {'Capital': 'Lusaka', 'GDP': '29 billion USD', 'Region': 'Southern Africa'},
'Mozambique': {'Capital': 'Maputo', 'GDP': '18 billion USD', 'Region': 'Southern Africa'},
'Botswana': {'Capital': 'Gaborone', 'GDP': '20 billion USD', 'Region': 'Southern Africa'},
'Namibia': {'Capital': 'Windhoek', 'GDP': '12 billion USD', 'Region': 'Southern Africa'},
'Malawi': {'Capital': 'Lilongwe', 'GDP': '13 billion USD', 'Region': 'Southern Africa'},
'Lesotho': {'Capital': 'Maseru', 'GDP': '3 billion USD', 'Region': 'Southern Africa'},
'Eswatini': {'Capital': 'Mbabane', 'GDP': '5 billion USD', 'Region': 'Southern Africa\n'},

# NORTH AFRICA
'Egypt': {'Capital': 'Cairo', 'GDP': '476 billion USD', 'Region': 'North Africa'},
'Morocco': {'Capital': 'Rabat', 'GDP': '142 billion USD', 'Region': 'North Africa'},

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# NORTH AFRICA
'Morocco': {'Capital': 'Rabat', 'GDP': '142 billion USD', 'Region': 'North Africa'},
'Algeria': {'Capital': 'Algiers', 'GDP': '239 billion USD', 'Region': 'North Africa'},
'Tunisia': {'Capital': 'Tunis', 'GDP': '46 billion USD', 'Region': 'North Africa'},
'Libya': {'Capital': 'Tripoli', 'GDP': '53 billion USD', 'Region': 'North Africa'},
'Sudan': {'Capital': 'Khartoum', 'GDP': '34 billion USD', 'Region': 'North Africa\n'},

# CENTRAL AFRICA
'Cameroon': {'Capital': 'Yaounde', 'GDP': '45 billion USD', 'Region': 'Central Africa'},
'DR Congo': {'Capital': 'Kinshasa', 'GDP': '65 billion USD', 'Region': 'Central Africa'},
'Congo': {'Capital': 'Brazzaville', 'GDP': '14 billion USD', 'Region': 'Central Africa'},
'Gabon': {'Capital': 'Libreville', 'GDP': '20 billion USD', 'Region': 'Central Africa'},
'Chad': {'Capital': 'N'Djamena', 'GDP': '12 billion USD', 'Region': 'Central Africa'},
'Angola': {'Capital': 'Luanda', 'GDP': '106 billion USD', 'Region': 'Central Africa'},
'Central African Republic': {'Capital': 'Bangui', 'GDP': '3 billion USD', 'Region': 'Central Africa'},
'Equatorial Guinea': {'Capital': 'Malabo', 'GDP': '12 billion USD', 'Region': 'Central Africa'},
'São Tomé and Príncipe': {'Capital': 'São Tomé', 'GDP': '1 billion USD', 'Region': 'Central Africa'},
}

for country, details in africa.items():
    print(f"Country: {country} | Capital: {details['Capital']} | GDP: {details['GDP']} | Region: {details['Region']}")

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Country: Nigeria | Capital: Abuja | GDP: 477 billion USD | Region: West Africa
Country: Ghana | Capital: Accra | GDP: 76 billion USD | Region: West Africa
Country: Senegal | Capital: Dakar | GDP: 31 billion USD | Region: West Africa
Country: Ivory Coast | Capital: Yamoussoukro | GDP: 70 billion USD | Region: West Africa
Country: Mali | Capital: Bamako | GDP: 19 billion USD | Region: West Africa
Country: Burkina Faso | Capital: Ouagadougou | GDP: 18 billion USD | Region: West Africa
Country: Niger | Capital: Niamey | GDP: 14 billion USD | Region: West Africa
Country: Guinea | Capital: Conakry | GDP: 20 billion USD | Region: West Africa
Country: Benin | Capital: Porto-Novo | GDP: 17 billion USD | Region: West Africa
Country: Togo | Capital: Lome | GDP: 9 billion USD | Region: West Africa
Country: Sierra Leone | Capital: Freetown | GDP: 4 billion USD | Region: West Africa
Country: Liberia | Capital: Monrovia | GDP: 4 billion USD | Region: West Africa
Country: Guinea-Bissau | Capital: Bissau | GDP: 2 billion USD | Region: West Africa
Country: Gambia | Capital: Banjul | GDP: 2 billion USD | Region: West Africa
Country: Cape Verde | Capital: Praia | GDP: 2 billion USD | Region: West Africa
Country: Mauritania | Capital: Nouakchott | GDP: 10 billion USD | Region: West Africa

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Country: Kenya | Capital: Nairobi | GDP: 118 billion USD | Region: East Africa
Country: Ethiopia | Capital: Addis Ababa | GDP: 156 billion USD | Region: East Africa
Country: Tanzania | Capital: Dodoma | GDP: 79 billion USD | Region: East Africa
Country: Uganda | Capital: Kampala | GDP: 49 billion USD | Region: East Africa
Country: Rwanda | Capital: Kigali | GDP: 14 billion USD | Region: East Africa
Country: Somalia | Capital: Mogadishu | GDP: 8 billion USD | Region: East Africa
Country: Burundi | Capital: Gitega | GDP: 3 billion USD | Region: East Africa
Country: Djibouti | Capital: Djibouti City | GDP: 4 billion USD | Region: East Africa
Country: Eritrea | Capital: Asmara | GDP: 2 billion USD | Region: East Africa
Country: South Sudan | Capital: Juba | GDP: 11 billion USD | Region: East Africa
Country: Madagascar | Capital: Antananarivo | GDP: 14 billion USD | Region: East Africa
Country: Comoros | Capital: Moroni | GDP: 1 billion USD | Region: East Africa
Country: Mauritius | Capital: Port Louis | GDP: 14 billion USD | Region: East Africa
Country: Seychelles | Capital: Victoria | GDP: 2 billion USD | Region: East Africa

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Country: South Africa | Capital: Pretoria | GDP: 405 billion USD | Region: Southern Africa
Country: Zimbabwe | Capital: Harare | GDP: 26 billion USD | Region: Southern Africa
Country: Zambia | Capital: Lusaka | GDP: 29 billion USD | Region: Southern Africa
Country: Mozambique | Capital: Maputo | GDP: 18 billion USD | Region: Southern Africa
Country: Botswana | Capital: Gaborone | GDP: 20 billion USD | Region: Southern Africa
Country: Namibia | Capital: Windhoek | GDP: 12 billion USD | Region: Southern Africa
Country: Malawi | Capital: Lilongwe | GDP: 13 billion USD | Region: Southern Africa
Country: Lesotho | Capital: Maseru | GDP: 3 billion USD | Region: Southern Africa
Country: Eswatini | Capital: Mbabane | GDP: 5 billion USD | Region: Southern Africa

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Country: Egypt | Capital: Cairo | GDP: 476 billion USD | Region: North Africa
Country: Morocco | Capital: Rabat | GDP: 142 billion USD | Region: North Africa
Country: Algeria | Capital: Algiers | GDP: 239 billion USD | Region: North Africa
Country: Tunisia | Capital: Tunis | GDP: 46 billion USD | Region: North Africa
Country: Libya | Capital: Tripoli | GDP: 53 billion USD | Region: North Africa
Country: Sudan | Capital: Khartoum | GDP: 34 billion USD | Region: North Africa

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Country: Cameroon | Capital: Yaounde | GDP: 45 billion USD | Region: Central Africa
Country: DR Congo | Capital: Kinshasa | GDP: 65 billion USD | Region: Central Africa
Country: Congo | Capital: Brazzaville | GDP: 14 billion USD | Region: Central Africa
Country: Gabon | Capital: Libreville | GDP: 20 billion USD | Region: Central Africa
Country: Chad | Capital: N'Djamena | GDP: 12 billion USD | Region: Central Africa
Country: Angola | Capital: Luanda | GDP: 106 billion USD | Region: Central Africa
Country: Central African Republic | Capital: Bangui | GDP: 3 billion USD | Region: Central Africa
Country: Equatorial Guinea | Capital: Malabo | GDP: 12 billion USD | Region: Central Africa
Country: São Tomé and Príncipe | Capital: São Tomé | GDP: 1 billion USD | Region: Central Africa

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